MILITARY SPECIFICATION SHEET

# CONNECTORS, ELECTRICAL, CIRCULAR, MINIATURE, COMPOSITE, HIGH DENSITY, QUICK COUPLING, ENVIRONMENT RESISTANT, CONTACTS, PIN, CRIMP REMOVABLE, (FOR MIL-C-38999 AND MIL-C-29600 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest revision of MIL-C-39029.

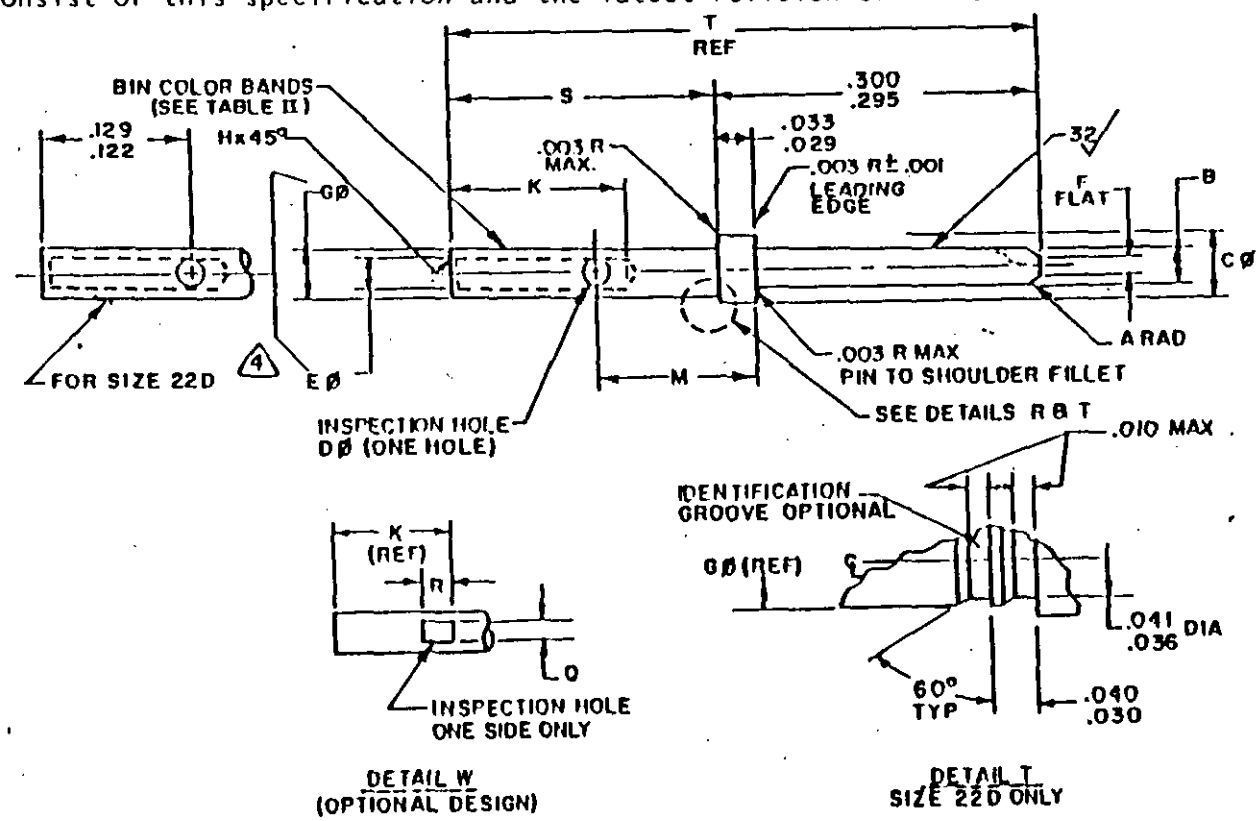



FIGURE 1. Connector contact.

NOTES:

1. Dimensions are in inches unless otherwise specified.
2. Metric equivalents are based upon 1 inch = 25.4 mm.
3. Dimensions apply after plating.
4. For BIN-614 only, diameters E and G shall be concentric within .003 full indicator movement (FIM) regardless of feature size (S); for all other contact sizes, diameters E and G to be concentric within .001 FIM of maximum material condition (M).
5. Manufacturers trademark per AIR-1351

AMSC N/A FSC 5935

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TABLE I. Dimensions.

| BIN Code | A Radius | BØ | CØ | DØ | EØ | FØ | GØ | H | K | M | Q | R | S | T Ref |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 620 | .020 | .0305 | .062 | .022 | .0355 | .011 Max | .048 | .005 | .157 | -- | .022 | .046 | .237 | .531 |
| | .010 | .0295 | .060 | .018 | 0.335 | | .046 | .003 | .141 | -- | .018 | .018 | .231 | |
| 621 | .025 | .041 | .094 | .032 | .048 | .015 Max | .070 | .010 | .229 | .078 | .032 | .063 | .237 | .531 |
| | .015 | .039 | .091 | .026 | .046 | | .068 | .005 | .209 | .072 | .026 | .026 | .231 | |
| 622 | .025 | .0635 | .130 | .042 | .068 | .030 | .103 | .010 | .229 | .088 | .042 | .073 | .237 | .531 |
| | .020 | .0616 | .127 | .036 | .066 | .011 | .101 | .005 | .209 | .082 | .030 | .036 | .231 | |
| 623 | .025 | .095 | .182 | .042 | .102 | .062 | .151 | .016 | .229 | .088 | .042 | .073 | .237 | .531 |
| | .020 | .093 | .179 | .036 | .098 | .043 | .148 | .005 | .209 | .082 | .030 | .036 | .231 | |
| 624 | .025 | .126 | .242 | .052 | .140 | .094 | .213 | .016 | .385 | .115 | N/A | N/A | .405 | .647 |
| | .020 | .124 | .238 | .040 | .134 | .074 | .207 | .005 | .355 | .108 | | | .395 | |

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .003 | .0762 | .041 | 1.041 | .091 | 2.311 | .157 | 3.988 |
| .005 | .1270 | .042 | 1.067 | .093 | 2.362 | .179 | 4.547 |
| .010 | .2540 | .043 | 1.092 | .094 | 2.388 | .182 | 4.623 |
| .011 | .2794 | .046 | 1.168 | .095 | 2.413 | .207 | 5.258 |
| .015 | .3810 | .048 | 1.219 | .098 | 2.489 | .209 | 5.309 |
| .016 | .4064 | .060 | 1.524 | .101 | 2.565 | .213 | 5.410 |
| .018 | .4572 | .0616 | 1.565 | .102 | 2.591 | .229 | 5.817 |
| .020 | .5080 | .062 | 1.575 | .103 | 2.616 | .231 | 5.867 |
| .022 | .5588 | .063 | 1.600 | .108 | 2.743 | .237 | 6.020 |
| .025 | .6350 | .0635 | 1.613 | .115 | 2.921 | .238 | 6.045 |
| .026 | .6604 | .066 | 1.676 | .124 | 3.150 | .242 | 6.147 |
| .0295 | .7493 | .068 | 1.727 | .126 | 3.200 | .252 | 6.401 |
| .030 | .7620 | .070 | 1.778 | .127 | 3.226 | .355 | 9.017 |
| .0305 | .7747 | .072 | 1.829 | .130 | 3.302 | .385 | 9.779 |
| .032 | .8128 | .073 | 1.854 | .134 | 3.404 | .395 | 10.033 |
| .0335 | .8509 | .074 | 1.880 | .140 | 3.556 | .405 | 10.287 |
| .0355 | .9017 | .078 | 1.981 | .141 | 3.581 | .531 | 13.49 |
| .036 | .9144 | .082 | 2.083 | .148 | 3.759 | .647 | 16.434 |
| .039 | .9906 | .088 | 2.235 | .151 | 3.835 | | |
| .040 | 1.016 | | | | | | |



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TABLE II. Design characteristics.

| BIN code | Color bands | | | Mating end size | Wire barrel size | Type | Class |
|---|---|---|---|---|---|---|---|
| | 1st | 2nd | 3rd | | | | |
| 620 | Blue | Red | Black | 22 | 22D | | |
| 621 | Blue | Red | Brown | 20 | 20 | | |
| 622 | Blue | Red | Red | 16 | 16 | A | B |
| 623 | Blue | Red | Orange | 12 | 12 | | |
| 624 | Blue | Red | Yellow | 10 | 10 | | |

TABLE III. Tools.

| Wire barrel size | Basic crimping tool | Positioner | Installing tool | Removal tool |
|---|---|---|---|---|
| 22D | M22520/2-01<br>M22520/7-01 | M22520/2-09<br>M22520/7-07 | M81969/14-01<br>M81969/8-01 | M81969/14-01<br>M81969/8-02 |
| 20 | M22520/1-01<br>M22520/2-01<br>M22520/7-01 | M22520/1-04<br>Red<br>M22520/2-10<br>M22520/7-08 | M81969/14-10<br>M81969/8-05 | M81969/14-10<br>M81969/8-06 |
| 16 | M22520/1-01<br>M22520/7-01 | M22520/1-04<br>Blue<br>M22520/7-04 | M81969/8-07<br>M81969/14-03 | M81969/8-08<br>M81969/14-03 |
| 12 | M22520/1-01 | M22520/1-04<br>Yellow | M81969/14-04<br>M81969/8-09 | M81969/14-04<br>M81969/8-10 |
| 10 | M22520/XX 1/ | M22520/XX 1/ | M81969/8-XX 1/<br>M81969/14-05 | M81969/8-YY 1/<br>M81969/14-05 |

1/ To be determined

REQUIREMENTS:

Dimensions: See Figure 1 and Table I.

Design characteristics and configuration: See Figure 1 and Table II.

Tools: See Table III.

Mating contacts: MIL-C-39029/106.

Temperature: -65°C to +175°C.



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REQUIREMENTS (continued):

Durability: 1500 cycles.

Random vibration: Connectors shall be subjected to the test specified in method 2005 of MIL-STD-1344. The following details shall apply:

a. Test condition V. Using the vibration envelope shown on figure 2. (Derived from zone 2 outlined in Aerospace Information Report AIR 1557.)
b. Vibration to be conducted at standard test conditions.
c. Duration shall be 8 hours in the longitudinal direction and 8 hours in a perpendicular direction for a total of 16 hours.

High-impact shock: Connectors shall be coupled together by normal coupling means. All contacts shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit during high-impact shock. Connectors shall be monitored for any discontinuities. A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used. Wired and mated connectors shall be subjected to the test specified in MIL-S-901, grade A with the following modifications and additions. Mounting fixture shall be in accordance with MIL-S-901, light weight. The cable or wire bundle shall be supported on a stationary frame in such a manner to provide a free flexing cable length between frame and fixture of not less than 36 inches (914.4 mm).

Test condition A. The plug shall be terminated with at least 80 percent of wired contacts. The wire bundle shall be provided with straight, open frame, strain felief accessory hardware.

Qualification: The qualification requirements shall consist of the sample numbers and tests specified in MIL-C-39029 for type A contacts with localized finish. The vibration test group shall consist of double the number of samples specified in MIL-C-39029. For those tests which require the use of a connector pair, either a composite or metal shell connector pair may be utilized. Caution should be used if exceeding the rated durability cycles of a metal shell connector. For vibration and shock testing, accessories shall be as specified in MIL-C-39029.

Periodic qualification: The manufacturer shall forward a report every 24 months to the qualifying activity. The periodic qualification requirements shall consist of those tests specified for qualification except for vibration and shock.

Quality conformance inspection: The quality conformance requirements shall consist of Groups A and B inspection as specified in MIL-C-39029.

Material: Base: Conductive copper alloy.
Plating: See 3.3.6 of MIL-C-39029.

Example of part number: M39029/107-618

- M39029/107: Specification sheet number
- 618: BIN code



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Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AV
Navy - EC
Air Force - 15
DLA - ES
NASA - NA

User activities:
Army - MI
Navy - SH, MC
Air Force - 15

Preparing activity:
Navy - AS
(Project No. 5935-3791-02)



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